Gyro Logic: A Stability-Based Framework for Representation Under Intrinsic Deviation

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Abstract

Gyro Logic is a theoretical framework for modeling representation under intrinsic deviation (Δ). In contrast to conventional approaches that assume observation as an approximation of reality, this framework treats deviation as an inherent and unavoidable structural component of representation.

Observation is formalized as a reconfiguration process (Slice), which transforms an underlying structure into a representation that necessarily includes deviation. Stability is defined as a function of deviation, representing the degree to which a representation remains consistent under perturbation.

Meaning emerges as stability-preserving regions under bounded deviation, while identity is defined as a trajectory that maintains stability over time. When deviation exceeds a critical threshold, representations collapse and undergo discontinuous transitions (Jump). Void is defined as the limit where deviation becomes unevaluable.

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1. Introduction

Gyro Logic proposes that reality is not directly observed, but constructed through observation.

The framework consists of three elements:

- Structure S

- Slice O
- Stability

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2. Core Architecture

$$X = O(S)$$

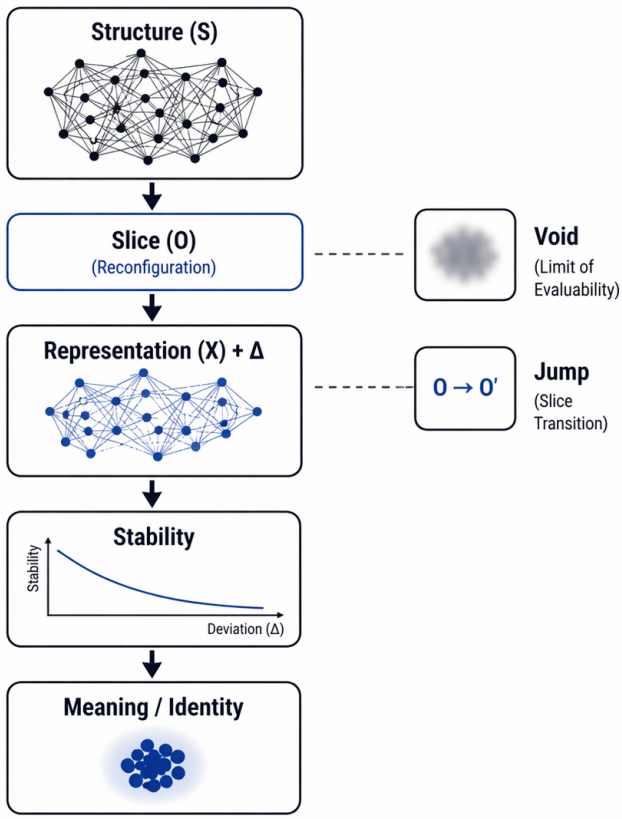

Figure 1. Core Architecture of Gyro Logic

Structure is reconfigured through Slice into a representation that inherently contains deviation (Δ). Stability evaluates whether the deviation remains within a tolerable range, determining the emergence of Meaning and Identity. Void represents the limit of evaluability, while Jump denotes a transition between slices.

Figure 1: Core Architecture of Gyro Logic

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3. Slice-Dependent Representation

$$X_1 = O_1(S), \quad X_2 = O_2(S)$$

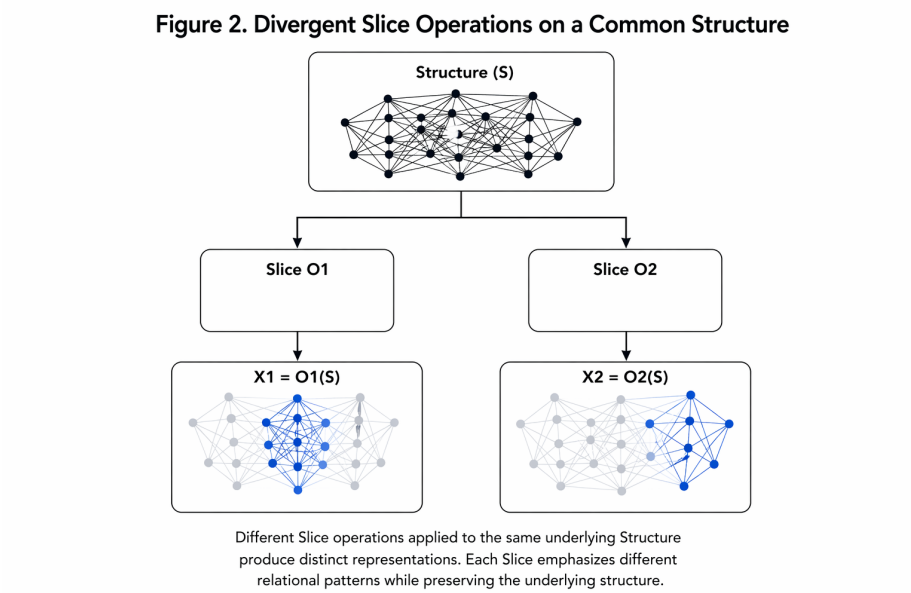

Figure 2: Divergent Slice Operations on a Common Structure

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4. Stability as a Function of Deviation

Let $\mathcal{S}$ be the structure space and $\mathcal{X}$ the representation space.

$$O : \mathcal{S} \to \mathcal{X}$$

$$d : \mathcal{X} \times \mathcal{X} \to \mathbb{R}_{\geq 0}$$

$$\Delta(S) := \mathbb{E}_{O_1, O_2}\left[d(O_1(S), O_2(S))\right]$$

$$\text{Stability}(S) := \exp(-\lambda \Delta(S)), \quad \lambda > 0$$

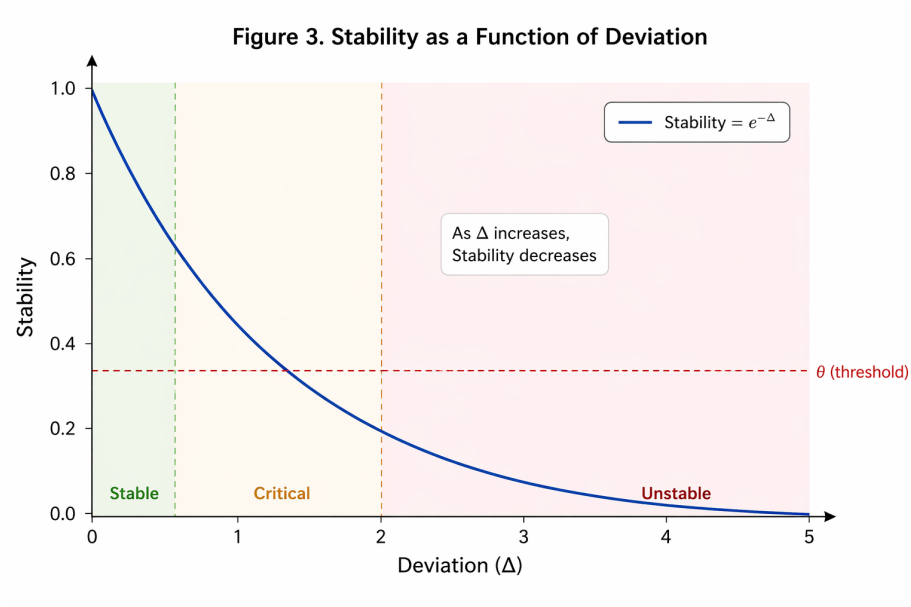

Figure 3: Stability as a Function of Deviation

5. Collapse into Void

$$\text{Void} := \{X \mid \Delta(X) \text{ undefined or diverges}\}$$

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6. Discontinuous Transition (Jump)

$$\Delta > \theta \Rightarrow O \to O'$$

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7. Identity as Stable Trajectory

$$\gamma : [0, T] \to \mathcal{X}$$

$$\mathcal{J}_\theta := \{\gamma \mid \text{Stability}(\gamma(t)) \geq \theta\}$$

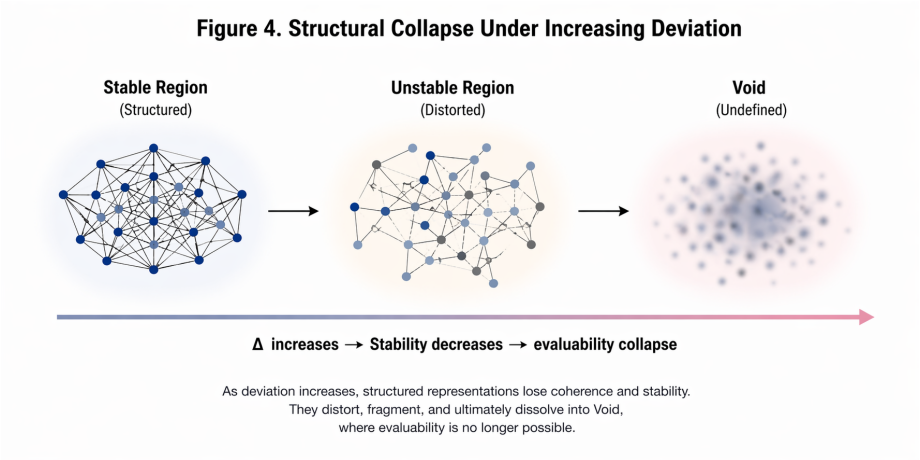

Figure 4: Structural Collapse Under Increasing Deviation

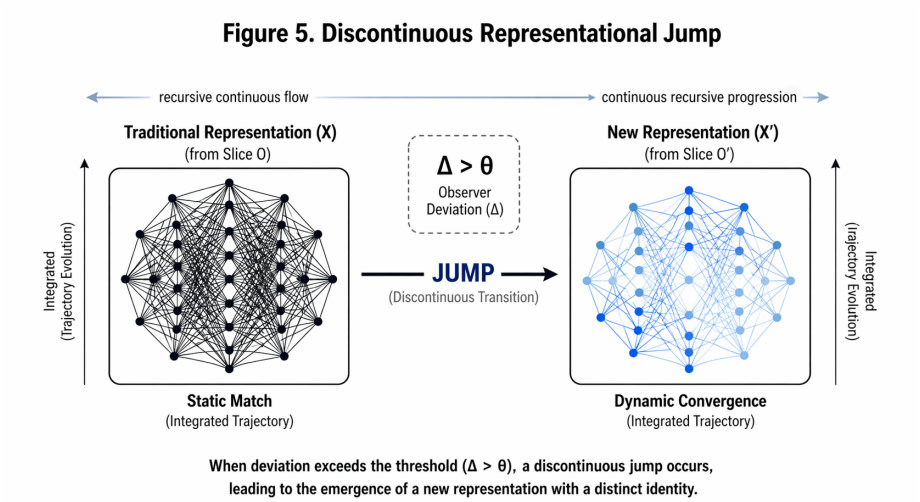

Figure 5: Discontinuous Representational Jump

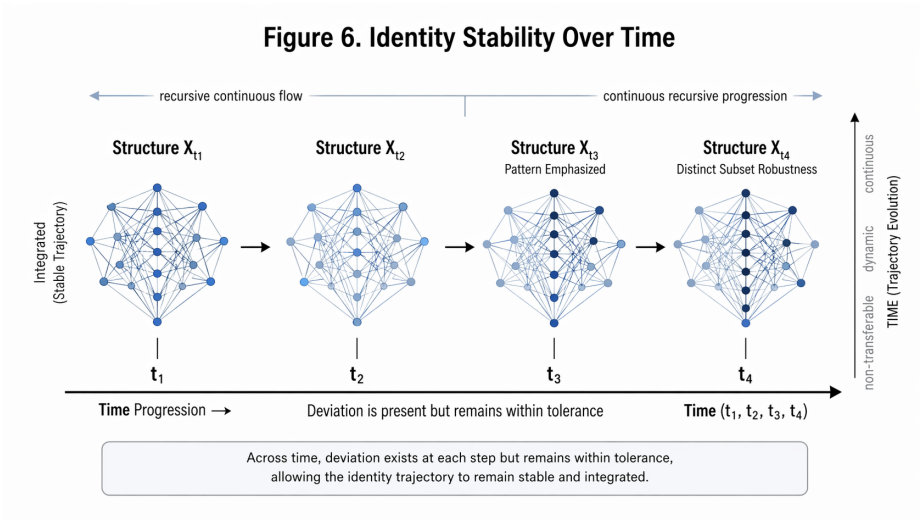

Figure 6: Identity Stability Over Time

8. Conclusion

Gyro Logic provides a unified framework for representation, deviation, and stability.

Appendix A. Formal Definitions

$$O : \mathcal{S} \to \mathcal{X}$$

$$X = O(S)$$

$$\Delta(S) := \mathbb{E}_{O_1, O_2}\left[d(O_1(S), O_2(S))\right]$$

$$\text{Stability}(S) := \exp(-\lambda \Delta(S))$$

$$\mathcal{M}_\theta := \{X \mid \text{Stability}(X) \geq \theta\}$$

$$\text{Void} := \{X \mid \Delta(X) \text{ undefined or diverges}\}$$

$$\mathcal{J}_\theta := \{\gamma \mid \text{Stability}(\gamma(t)) \geq \theta\}$$

Appendix B. Theoretical Results

$$0 < \text{Stability}(S) \leq 1$$

$$\Delta_1 < \Delta_2 \Rightarrow \text{Stability}_1 > \text{Stability}_2$$

$$X \in \mathcal{M}_\theta \iff \text{Stability}(X) \geq \theta$$

$$\Delta(X) \to \infty \Rightarrow \text{Stability}(X) \to 0$$